Carboxylase and PEP Carboxylase) and involved in the synthesis of DNA and RNA. It is essential for binding of ribosomal sub units. It is absorbed as Mg^{2+} ions.

Deficiency symptoms: Inter veinal chlorosis, necrosis, anthocyanin (purple) formation and Sand drown of tobacco.

6. **Sulphur (S):** Essential component of amino acids like cystine, cysteine and methionine, constituent of coenzyme A, Vitamins like biotin and thiamine, constituent of proteins and ferredoxin. plants utilise sulphur as sulphate (SO_4^-) ions.

   Deficiency symptoms: Chlorosis, anthocyanin formation, stunted growth, rolling of leaf tip and reduced nodulation in legumes.



NPK Fertilizers

It consists of nitrogen, phosphate with potassium in different proportions. The number labelled on the bags as 15:15:15 indicates N, P & K in equal proportions.

12.3 Functions, mode of absorption and deficiency symptoms of micronutrients

Micronutrients even though required in trace amounts are essential for the metabolism of plants. They play key roles in many plants. Example: Boron is essential for translocation of sugars, molybdenum is involved in nitrogen metabolism and zinc is needed for biosynthesis of auxin. Here, we will study about the role of micro nutrients, their functions, their mode of absorption, deficiency symptoms and deficiency diseases.

1. **Iron (Fe):** Iron is required lesser than macronutrient and larger than micronutrients, hence, it can be placed in any one of the groups. Iron is an essential element for the synthesis of chlorophyll and carotenoids. It is the component of cytochrome, ferredoxin, flavoprotein, formation of chlorophyll, porphyrin, activation of catalase, peroxidase enzymes. It is absorbed as ferrous (Fe^{2+}) and ferric (Fe^{3+}) ions. Absorbtion of Fe^{2+} ions are comparitively more than Fe^{3+} ions. Mostly fruit trees are sensitive to iron.

   Deficiency symptoms : Interveinal Chlorosis, formation of short and slender stalk and inhibition of chlorophyll formation.

2. **Manganese (Mn):** Activator of carboxylases, oxidases, dehydrogenases and kinases, involved in splitting of water to liberate oxygen (photolysis). It is absorbed as manganous (Mn^{2+}) ions.

   Deficiency symptoms : Interveinal chlorosis, grey spot on oats leaves and poor root system.

3. **Copper (Cu):** Constituent of plastocyanin, component of phenolases, tyrosinase, enzymes involved in redox reactions, synthesis of ascorbic acid, maintains carbohydrate and nitrogen balance, part of oxidase and cytochrome oxidase. It is absorbed as cupric (Cu^{2+}) ions.

   Deficiency symptoms : Die back of citrus, Reclamation disease of cereals and legumes, chlorosis, necrosis and Exanthema in *Citrus*.

4. **Zinc (Zn):** Essential for the synthesis of Indole acetic acid (Auxin), activator of carboxylases, alcohol dehydrogenase, lactic dehydrogenase, glutamic acid dehydrogenase, carboxy peptidases and tryptophan synthetase. It is absorbed as Zn^{2+} ions.

   Deficiency symptoms : Little leaf and mottle leaf due to deficiency of auxin, Inter veinal chlorosis, stunted growth, necrosis and Khaira disease of rice.

XIth_Botany_Chapter_12_EM.indd 262 05-01-2023 18:29:32